CONTIGUOUS MEMORY ALLOCATION
OPERATING SYSTEMS CONCEPT – LAB EXP – 09:

FIRST FIT

Activities Terminal Apr 20 17:07

snoordeen@localhost:~/os-lab/firstfit — vim firstfit.c

```c
#include <stdio.h>
int main(){
     int i;
     printf("Enter number of memory blocks:\n");
     scanf("%d", &i);
     int blocksSize[i], remainingSize[i];
     printf("Enter size of each block:\n");
     for(int i=0; i<i; i++){
          scanf("%d", &blocksSize[i]);
          remainingSize[i]=blocksSize[i];
     }
     printf("Enter number of processes:\n");
     scanf("%d", &i);
     int processSize[i];
     printf("Enter size of each process:\n");
     for(int i=0;i<i; i++){
          scanf("%d", &processSize[i]);
     }
     for(int i=0; i<i; i++){
          int allocated = 0;
          for(int k=0; i<i; k++){
               if(remainingSize[k] >= processSize[i]){
                    int fragment = remainingSize[k] - processSize[i];
                    printf("Process %d of size %d is allocated to Block %d of size %d with fragment %d\n",
                         i+1, processSize[i], k+1, blocksSize[k], fragment);
                   remainingSize[k] = processSize[i];
                   allocated = 1;
                   break;
               }
          }
          if(!allocated){
               printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
          }
     }
     return 0;
}
```

1,1 All

Activities Terminal Apr 20 17:09

snoordeen@localhost:~/os-lab/firstfit

```
snoordeen@localhost firstfit$ vi firstfit.c
snoordeen@localhost firstfit$ gcc firstfit.c -o firstfit
snoordeen@localhost firstfit$ ./firstfit
Enter number of memory blocks:
3
Enter size of each block:
100
200
300
Enter number of processes:
3
Enter size of each process:
212
112
Process 1 of size 212 is allocated to Block 2 of size 200 with fragment 188
Process 2 of size 417 is allocated to Block 3 of size 300 with fragment 183
Process 3 of size 112 is allocated to Block 2 of size 200 with fragment 176
snoordeen@localhost firstfit$
```

CONTIGUOUS MEMORY ALLOCATION
OPERATING SYSTEMS CONCEPT – LAB EXP – 09:

BEST FIT

Activities   Terminal   Apr 20 17:04

snoordeen@localhost:~/os-lab/bestfit — vim bestfit.c

```c
#include <stdio.h>
int main(){
    int i, j;
    printf("Enter number of memory blocks:\n");
    scanf("%d", &i);
    int blocksSize[i], remainingSize[i];
    printf("Enter size of each block:\n");
    for(int i=0; i<i; i++){
        scanf("%d", &blocksSize[i];
        remainngSize[i]=blocksSize[i];
    }
    printf("Enter number of processes:\n");
    scanf("%d", &j);
    int processSize[i];
    printf("Enter size of each process:\n");
    for(int i=0; i<j; i++){
        scanf("%d", &processSize[i]);
    }
    for(int i=0; i<j; i++){
        int bestIndex = -1;
        for(int k=0; k<i; k++){
            if(remainingSize[k]>= processSize[i]){
                if(bestIndex==-1 || remainingSize[k]<remainingSize[bestIndex]){
                    bestIndex=k;
                }
            }
        }
        if(bestIndex!=-1){
            int fragment=remainingSize[bestIndex]-processSize[i];
            printf("Process %d of size %d is allocated to Block %d of size %d with fragment %d\n", i+1, processSize[i], bestIndex+1,
                blocksIndex+,bestIndex], fragment);
            remainingSize[bestIndex]=processSize[i];
        } else{
            printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
        }
    }
    return 0;
}
```

37,1 All

Activities   Terminal   Apr 20 17:06

snoordeen@localhost:~/os-lab/bestfit

```
snoordeen@localhost:~/os-lab/bestfit$ vi bestfit.c
snoordeen@localhost:~/os-lab/bestfit$ gcc bestfit.c -o bestfit
snoordeen@localhost:~/os-lab/bestfit$ ./bestfit
Enter number of memory blocks:
3
Enter size of each block:
100
300
600
Enter number of processes:
3
Enter size of each process:
212
117
112
Process 1 of size 212 is allocated to Block 4 of size 600 with Fragment 48
Process 2 of size 117 is allocated to Block 2 of size 300 with Fragment 183
Process 3 of size 112 is allocated to Block 2 of size 300 with Fragment 176
snoordeen@localhost:~/os-lab/bestfit$
```